Forecast2Action AI

Hydroclimatic Risk & Early Warning System — Ethiopia

TECHNICAL BRIEF
IGAD HACKATHON 2026

Seasonal forecasts already know a drought or flood is coming weeks out — but that signal rarely reaches communities as something actionable, and it is almost never checked against what is actually happening on the ground. Forecast2Action AI closes that loop: real hazard data in, verified evidence in the loop, real action out.

Pipeline

Real Data	→	Risk Scoring	→	Staged AI	→	Community Loop	→	Action Output
WorldPop · GLW4 · FEWS NET · ECMWF SEAS5		P×S×E×V per hazard, per admin unit		Evidence → Synthesis → Action		Field reports, verified & weighted		Advisory · SMS · WhatsApp

Drought and wet-hazard risk are scored independently per region/zone/woreda as a relative 0–100 index:

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Risk = 100 × Probability × Severity × Exposure × Vulnerability
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Trigger / Warning / Watch / No-alert thresholds are calibrated against each hazard's own real observed ceiling for the current period (not a fixed theoretical maximum) — the worst nationally-observed drought risk score is only ~54/100, so a fixed threshold would make "Trigger" mathematically unreachable.

Real data, not synthetic

SIGNAL	SOURCE
Population & exposure	WorldPop 2020, 1 km
Livestock exposure	GLW4 (cattle)
Vulnerability	FEWS NET IPC food security
Seasonal hazard forecast	ECMWF SEAS5, 25-member ensemble
Admin boundaries	Region → zone → woreda, real polygons

Deterministic grounding

- The LLM **narrates**; every cited number is merged in afterward from computed evidence, never generated by the model.
- An automated validator flags any quoted figure that does not match the real underlying data — visible to the user, not hidden.
- Multi-provider fallback chain (Gemini, NVIDIA NIM, OpenRouter, OpenAI, Groq), tuned around real per-provider context-window, image-count, and rate-limit constraints found in testing.

Three-stage AI interpretation

- **Stage 1 — Evidence:** area-agnostic, layer-by-layer and indicator-by-indicator interpretation of hazard, exposure, and vulnerability rasters.
- **Stage 2 — Synthesis:** integrates cross-indicator findings into ranked priority areas, now including real community ground-truth evidence per area.
- **Stage 3 — Action:** translates synthesis into farmer, agro-pastoral, and humanitarian advisories, plus a GSM-7-aware SMS/WhatsApp summary.

Community verification loop

- Field reports (community focal points, extension workers, kebele/woreda officials, NGO partners) are submitted in real time — never cached stale.
- A verified report counts double toward the area's "ground signal" strength, so verification status changes the computed evidence, not just a display badge.
- The AI cites specific reports by count and type as corroborating — never proof — evidence.

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